



2022 3rd International Conference on Power, Energy and Electrical Engineering (PEEE 2022)
18–20 November 2022

State of health estimation for lithium battery random charging process based on CNN-GRU method

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Received 15 December 2022; accepted 27 December 2022

Available online 2 January 2023

Dataset link: <https://www.nasa.gov/content/prognostics-center-of-excellence-data-set-repository/#batteries>

Abstract

The accurate estimation of lithium battery state of health (SOH) is very important for the safe and stable operation of the battery. Since the user's charging process is random, it is difficult for the user to know the battery SOH through the charging segment. In this article, we proposed a lithium battery SOH estimation method of random charging process based on convolutional gated recurrent unit (CNN-GRU). The method extracts key features adaptively from the segments of voltage, current and temperature curves in the charging process through the CNN-GRU framework to realize the lithium battery SOH estimation. Compared with traditional methods, this method does not need to manually select or construct feature information and it can achieve high precision SOH evaluation. Through experimental verification, the error of this method can reach to 0.901%.

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Peer-review under responsibility of the scientific committee of the 3rd International Conference on Power, Energy and Electrical Engineering, PEEE, 2022.

Keywords: Random charging process; CNN-GRU; SOH; Lithium battery

1. Introduction

With the intensification of the global energy crisis, lithium battery, as a high-energy density and low-price energy storage device, has been widely used in electric vehicles, photovoltaic energy storage and other fields [1]. The self-aging of the battery and some man-made operation problems will cause the degradation of battery performance, the reduction of the battery's capacity, the possibility of short circuits, and even accidents such as fires [2]. The battery management system (BMS) [3,4] can detect the actual use of the battery, timely feedback and issue early warnings, so as to ensure the long-term safe and stable operation of the battery. Battery state of health (SOH) [5,6] and the remaining useful life of the battery (RUL) [7] are important indicators in BMS to characterize the degree of battery

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<https://doi.org/10.1016/j.egyr.2022.12.093>

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aging, the former is to estimate the current health of the battery, and the latter is to predict the future service life of the battery. This article focuses on the battery SOH estimation.

Battery SOH is defined as the ratio of the current capacity of the battery to the rated capacity [8]. As the number of charge and discharge times of the battery increases, the impedance inside the battery increases, and the storage capacity of the battery decreases, thereby reducing the SOH of the battery [9]. At present, the methods for battery SOH estimation mainly include model-based methods and data-driven methods [10]. Model-based methods construct corresponding battery models through the physicochemical reactions inside the battery, which are usually complex and computationally intensive, and the corresponding models need to be adjusted for different batteries [11].

The data-driven methods construct the connection between input features and battery SOH through the historical charge and discharge data of the battery. In recent years, with the rapid development of artificial intelligence [12], many data-driven methods have been applied to battery SOH estimation. Yang, NK. used CNN to extract SOH changes between charge–discharge cycles, and predicted SOH through random forest algorithm and CNN indicators [13]. Wen, JP. used BP neural network for SOH estimation through the analysis of the incremental capacity of the battery [14]. Li, QL used IALO-SVR algorithm to train the four features such as charging time proposed in the charging data, and accurately estimated the SOH of the battery [15]. Deng, ZW used a sparse Gaussian regression algorithm and an incremental sequence of random capacity to estimate the battery SOH [16].

At present, few people consider the random situation of users charging lithium batteries, and most researchers are experimenting with the entire charging process. They usually estimate the battery SOH by artificially extracting appropriate features from the entire charging data. In this article, a SOH estimation method for the random charging process of lithium batteries based on convolutional gated recurrent unit (CNN-GRU) is proposed. We first obtain the data of the random charging process of lithium batteries, and then process the features in the time and space dimensions, and finally realize the battery SOH estimation. The rest of this article is as follows: the second part introduces the process of battery data analysis and feature extraction; The third part explains the principle of the CNN-GRU algorithm; The fourth part is the experimental results of battery SOH estimation; Finally, summarizes the article.

2. Battery data analysis and feature extraction

2.1. Data analysis

The dataset used in this article is from the NASA PCOE Research Center [17]. The battery dataset records detailed data on current, voltage, and temperature in charge and discharge experiments of lithium batteries at different temperatures. In this article, battery B05 is selected as the experimental object, and the rated capacity of the battery is 2 Ah. Charge and discharge test at room temperature of 24 degrees Celsius: first charge in constant current (CC) mode of 1.5 A until the voltage reaches to 4.2 V, then hold this voltage for constant voltage (CV) mode charging until the charging current drops to 20 mA. In discharge mode, a constant current discharge is performed at a current of 2 A until the battery voltage drops to 2.7 V. In order to test the generalization of the model, B07 battery was added for control experiments, and unlike B05 batteries, the B07 battery is discharged with a cutoff voltage of 2.2 V. In this article, the first 136 charge and discharge cycle data of B05 battery and B07 battery are selected for experiments, and the actual capacity of the battery changes with the number of charges and discharges are shown in Fig. 1.

2.2. Feature extraction

The SOH estimate of a lithium battery is usually defined as the ratio of the current capacity to the rated capacity, as shown in Eq. (1).

$$SOH = \frac{C_i}{C_o} \times 100\% \quad (1)$$

where C_i represents the battery capacity of the i th cycle, and C_o represents the rated capacity. The selection of features plays a very important role in the accuracy of machine learning methods, and selecting appropriate features and algorithms can significantly improve the estimation effect of the model. Since the discharge situation of the battery in the actual situation is very complex and difficult to count, and the charging process is stable and the data

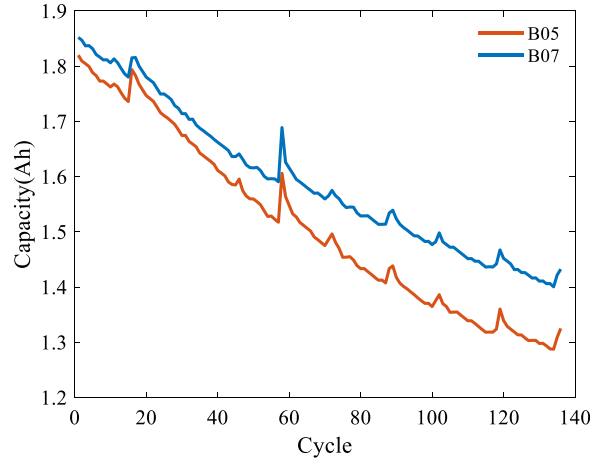


Fig. 1. Lithium battery capacity decay curve.

is easy to obtain, this article selects the charging data of the B05 battery as the input. The charging curves of B05 batteries under different SOH conditions are as follows.

We can find that the voltage, current, and temperature of different SOH at the same charging time are obviously different, and the time series of the three variables are also very different throughout the charging cycle in Fig. 2. This shows that the charging time series of voltage, current, and temperature can be used as the features to estimate the battery SOH.

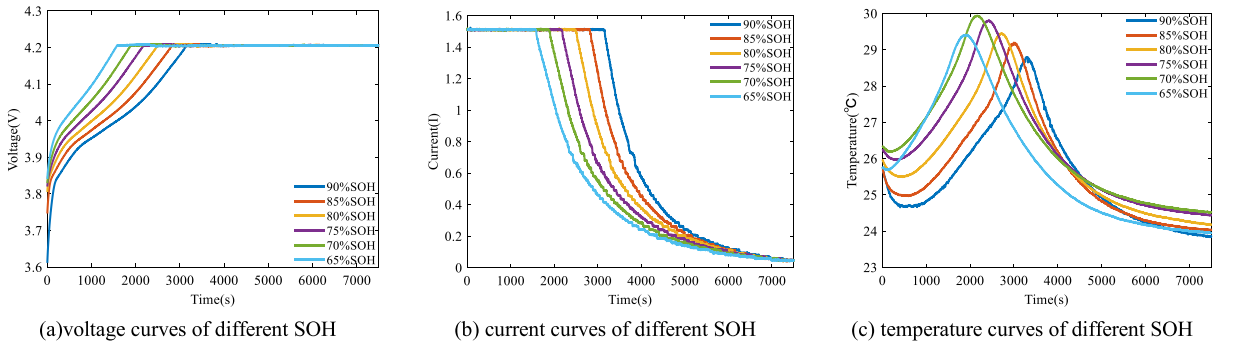


Fig. 2. Charging curves of different SOH.

Fig. 2 shows that the data dimensions of the three sequence features are quite different, in order to better carry out the experiment, the three sequence feature data are normalized, and the normalized expression is:

$$y' = \frac{y - y_{\min}}{y_{\max} - y_{\min}} \quad (2)$$

where y' is the value after normalization, y is the value before normalization, $y_{\max}$ is the sample maximum, and $y_{\min}$ is the sample minimum. Then the SOH of the battery can be expressed as:

$$SOH = f(S_v, S_i, S_t) \quad (3)$$

where S_v , S_i , S_t represent the time series of voltage, current, and temperature during charging. In order to simulate the randomness of the user's charging process, this article takes a total of 3000 sampling points per charging cycle (sampled every 2.5 s), and takes the first 2000 sampling points as a sample input. Then we set the step size to 10, then continuously obtain the sample according to this step. So each charging cycle can obtain 100 samples of different charging process, and these samples characterize information about the user's random charging process.

3. CNN-GRU method

3.1. Convolutional neural network

CNN is a typical artificial neural network suitable for processing spatial data. As shown in Fig. 3, CNN is composed of a convolutional layer, a pooling layer and a fully connected layer, the convolutional layer can extract the local features of the two-dimensional data; the pooling layer can reduce the dimension of the middle-hidden layer without damaging the features, reduce the amount of operations in the future; and the fully connected layer is classified by the features extracted by the convolution and pooling layer.

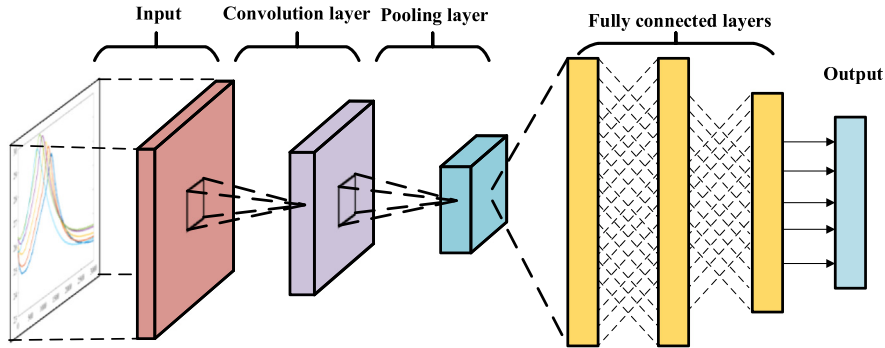


Fig. 3. CNN structure diagram.

3.2. Gate recurrent unit

GRU is a type of recurrent neural network, and compared to LSTM, GRU internally combines forgetting gates and input gates into an update memory stage. This makes the GRU less a gate, with fewer parameters than LSTM, which can greatly improve training efficiency.

The internal structure of the GRU is shown in Fig. 4. The four important parameters and the expressions of all parameters are shown below:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (4)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (5)$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t]) \quad (6)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (7)$$

where x_t is the input information of the current moment, z_t is the current update gating signal, r_t is the current reset gating signal, $\tilde{h}_t$ remembers the current input and part of the output of the previous moment, h_t represents the current output, and W_z , W_r , W represents the weight matrices, σ , $\tanh$ represent the activation functions.

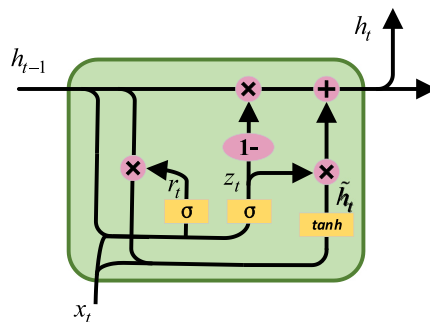


Fig. 4. GRU structure diagram.

3.3. CNN-GRU

CNN-GRU combines the advantages of GRU and CNN. On the one hand, the powerful feature extraction ability of CNN is used to mine the internal connections between data, thereby reducing the scale and complexity of the original data. On the other hand, through the timing memory ability of GRU neural network, the internal dynamic change law of the data is learned, and the nonlinear relationship between input and output is established. The CNN-GRU network constructed in this article consists of 4 convolution layers, 4 pooling layers, 1 GRU layers, and 2 fully connected layers, as shown in Fig. 5.

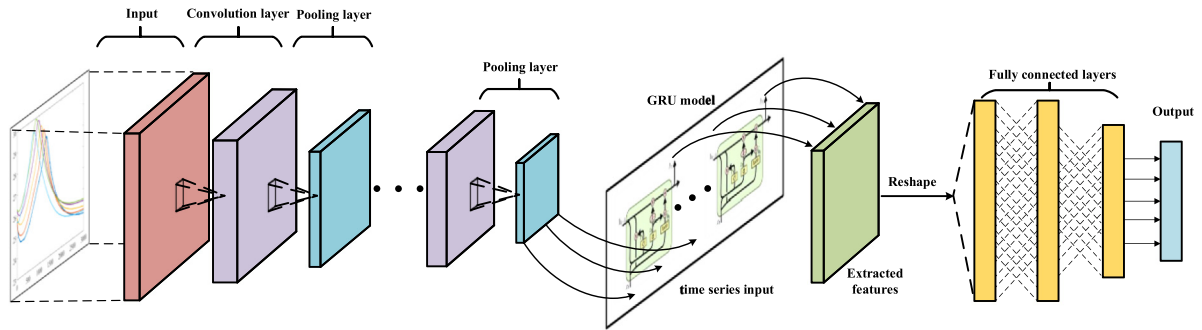


Fig. 5. CNN-GRU structure diagram.

After inputting the processed data into the model, the features are extracted through convolution operation, and then the data dimension is reduced by pooling operation. And we use the GRU layer to learn the time series relationship between the dimensionality reduction data, finally the features are converted into the final output result through the fully connected layer.

4. Experimental design and results analysis

This experiment uses the first 100 cycles of the B05 battery for training and the last 36 cycles for testing. The predicted battery SOH is shown in Fig. 6. We can find that the predict values of different charging process can well fit the real values, but there are still some fluctuations.

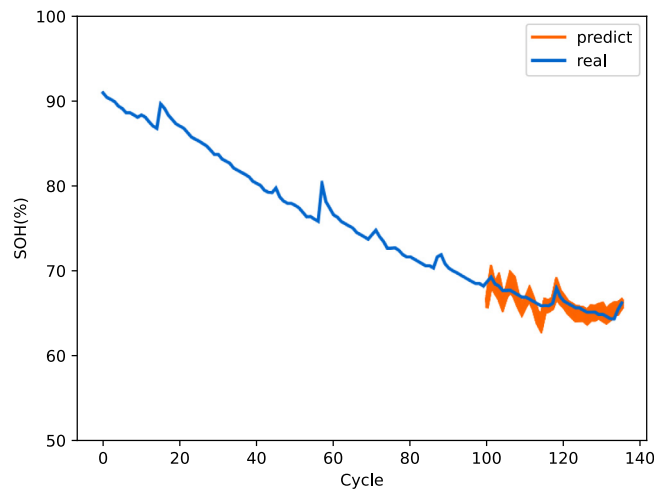


Fig. 6. B05 battery SOH prediction result.

In order to analyze the estimated performance of different methods, we use the mean absolute error (MAE) and the mean absolute percentage error (MAPE) to evaluate the estimated performance. This standard is defined in

(8), (9):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y'_i| \quad (8)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - y'_i}{y_i} \right| \times 100\% \quad (9)$$

where n is the number of samples, y_i is the real value and y'_i is the predicted value. In order to verify the effectiveness of the proposed method, CNN, Gaussian process regression (GP), sparse Gaussian process regression (SGP), and Back Propagation neural network (BP) were used for controlled experiments.

Fig. 7(a) shows the prediction results of CNN-GRU and CNN, and we can find that their fluctuations are similar. But the prediction results of CNN fluctuate more obviously, because CNN cannot get the time features. Fig. 7(b) and (c) show the comparison between GP, SGP and CNN-GRU respectively. We can find there are many outliers and fluctuations in the prediction results of GP method. And the results of SGP can well fit the real values, but the prediction range is too large. Finally, Fig. 7(d) shows that the prediction results of BP have a large deviation on the whole.

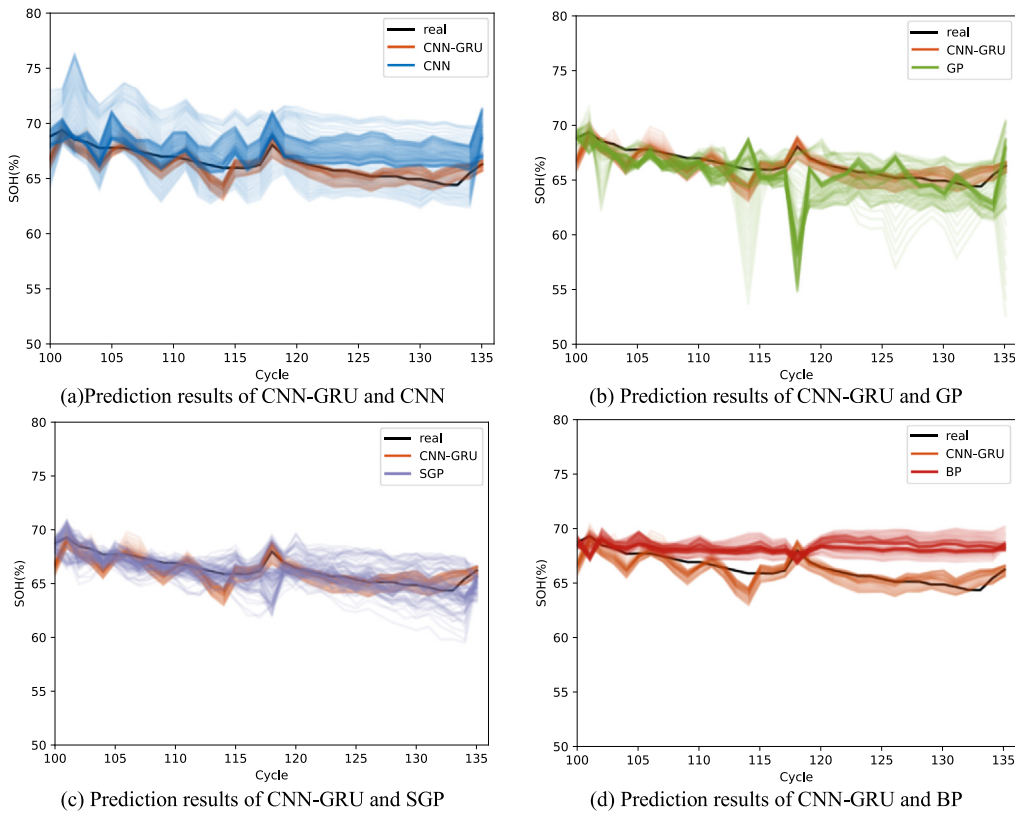


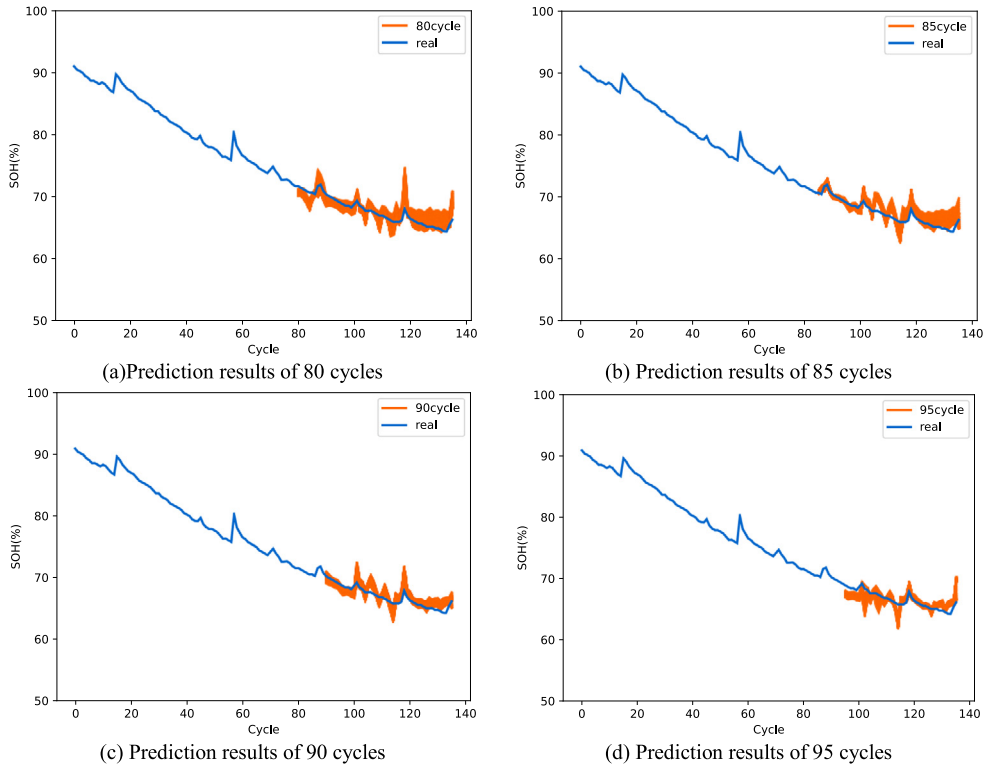
Fig. 7. Prediction results of different algorithms.

Table 1 shows the errors of the different methods. As the results of the previous analysis, we can find that the prediction error of BP is the largest, reaching to more than 2.04%. The prediction error of CNN is slightly smaller, down to 1.7%. The prediction error of SGP is lower than that of GP, down to 1.24%. The above methods do not focus on the time features of the random charging process, and CNN-GRU adaptively extracts time and space features, and the final prediction error reaches to 0.90%.

To verify the sensitivity of the CNN-GRU method, the training set was shortened to 95, 90, 85, and 80 cycles, and the results were as follows. Fig. 8(a), (b), (c) and (d) represent the predicted results of B05 battery with 80,

Table 1. Errors of different methods.

Prediction method	MAE	MAPE
BP	0.0204068	0.0310355
CNN	0.0170091	0.0257094
GP	0.0136866	0.0205926
SGP	0.0124975	0.0188328
CNN-GRU	0.0090119	0.0135967

**Fig. 8.** Prediction results of different cycles.

85, 90 and 95 cycles of test set respectively. As the test set shrinks, we can find that the range of prediction results becomes larger and the fluctuations become more obvious, especially at the inflection point of the SOH curve. We can find that when the number of cycles is 120, the smaller the test set, the worse the training result. Especially when the training set is 80 cycles, the error of this point can reach about 5%. It shows that the experimental results are reasonable, and the results are not too random.

Fig. 9 shows the prediction error of different methods in different cycles. We can find that the prediction errors of all methods are gradually decreasing, although the BP method has a little fluctuation. And the prediction error of CNN-GRU is the minimum at all stages. When the training set is 80 cycles, the error can also reach to 1.20%, even less than the prediction error of 1.24% when SGP is 100 cycles. It verifies the effectiveness of the proposed method.

In order to verify the generalization of the proposed method, B07 battery was used for detection, and CNN, SGP, BP methods were used for comparison. Fig. 10(a), (b), (c) and (d) represent the B07 battery predicted results of CNN-GRU, CNN, BP and SGP respectively. From the four figures, we can find the prediction results of CNN-GRU can fit the real values best. The prediction results of BP and CNN have obvious deviation in the previous part, even the prediction results of BP have some obvious outliers. The prediction results of SGP is the worst among the four methods, probably because this method cannot adapt to the change of time features in the random charging process of different batteries.

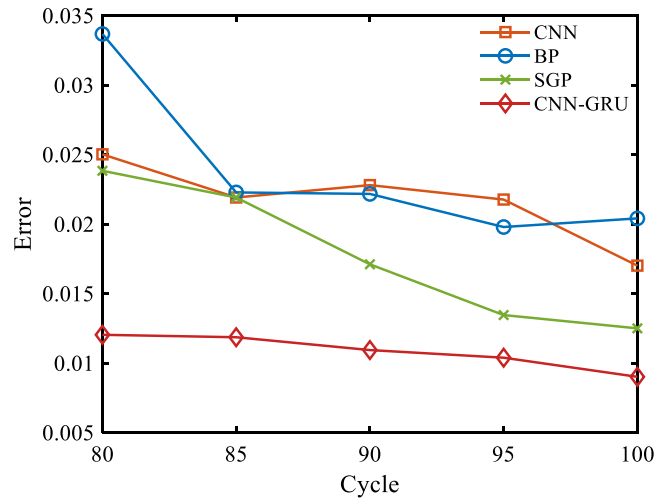


Fig. 9. Prediction results of different cycles.

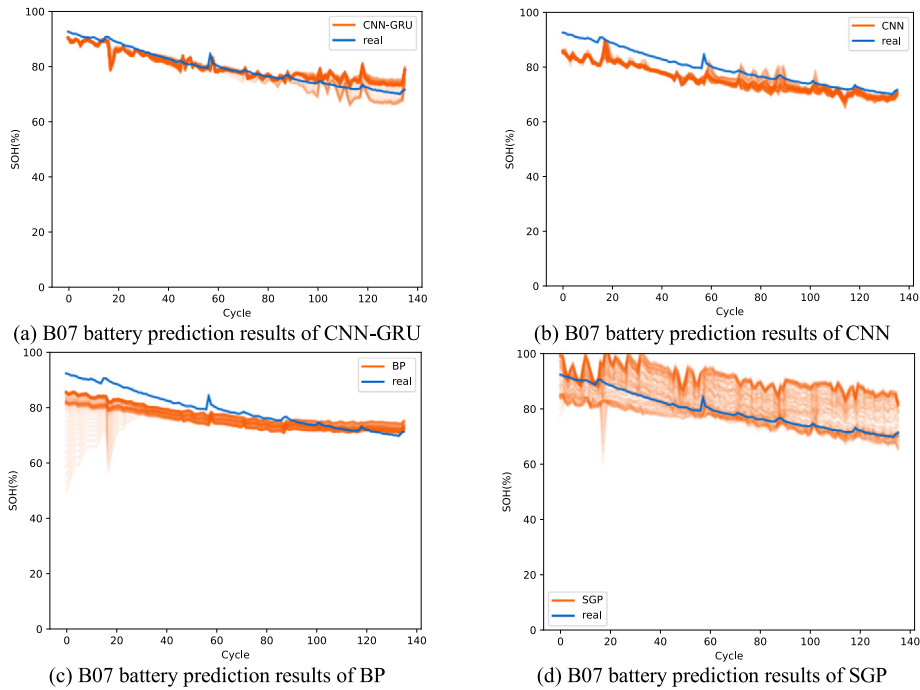


Fig. 10. Prediction results of different algorithms on B07 battery.

Table 2 shows the errors of different methods on B07 battery. As previously analyzed, the prediction error of SGP is the largest, reaching to 6.26%. The prediction errors of BP and CNN reach to 4.18% and 3.94%. The method proposed in this article has the best effect, and the prediction error can reduce to 1.94%. It shows that the method proposed in this article has good generalization and can be applied to other batteries.

5. Conclusion

In this article, we proposed a SOH estimation for lithium battery random charging process based on CNN-GRU. Firstly, we extract key features adaptively from the segments of current, voltage and temperature curves in the

Table 2. Errors of different methods on B07 battery.

Prediction method	MAE	MAPE
SGP	0.0626495	0.0800521
BP	0.0418851	0.0511886
CNN	0.0394435	0.0473144
CNN-GRU	0.0194312	0.0248129

charging process through the CNN-GRU framework to realize the lithium battery SOH estimation. Compared with CNN, BP, GP and SGP methods, the proposed method has the lowest prediction error, which can reach to 0.901%. In addition, the sensitivity experiment of different training set sizes proves that the prediction results of this method are reasonable and not too random. In the results of the generalization experiment on the B07 battery, the proposed method still has better performance than CNN, BP and SGP, but there are also some outliers. In future research, we will further consider the features of multiple batteries and construct a battery SOH estimation model with stronger generalization ability to achieve more accurate and reliable prediction results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are openly available at NASA PCOE Research Center at <https://www.nasa.gov/content/prognostics-center-of-excellence-data-set-repository/#batteries>.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grant 52107074.

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